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Operations Research & Optimization

2026 7 31

1. (Linear Programming)
2. (Nonlinear Programming)
3. (Heuristics & Metaheuristics)
4. (Convex Optimization)
5. ML
6. GPU / CUDA ML

(LP)

-
- /
- LP

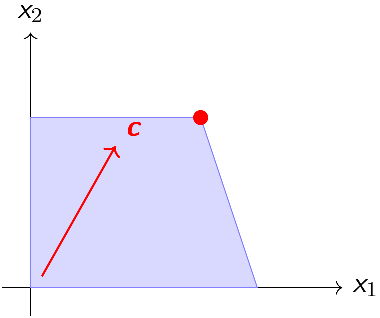
(Standard Form)

$$
\begin{aligned}
\min_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\
\text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\
& \boldsymbol{x} \geq \mathbf{0}
\end{aligned}
$$

$\boldsymbol{A} \in \mathbb{R}^{m \times n}$ $\boldsymbol{b} \in \mathbb{R}^{m}$ $\boldsymbol{c} \in \mathbb{R}^{n}$

LP

- **(Polyhedron)** $\{\boldsymbol{x} \mid \boldsymbol{Ax} = \boldsymbol{b},\ \boldsymbol{x} \geq 0\}$
- (Convex Polytope)
- **(extreme point / basic feasible solution)**
- (Simplex Method)



LP

LP

- $\max \boldsymbol{c}^\top \boldsymbol{x} \iff \min -\boldsymbol{c}^\top \boldsymbol{x}$
- $\boldsymbol{Ax} \le \boldsymbol{b} \iff \boldsymbol{Ax} + \boldsymbol{s} = \boldsymbol{b},\ \boldsymbol{s} \ge \boldsymbol{0}$
- $\boldsymbol{Ax} \ge \boldsymbol{b} \iff \boldsymbol{Ax} - \boldsymbol{s} = \boldsymbol{b},\ \boldsymbol{s} \ge \boldsymbol{0}$
- $x_i \quad \iff x_i = x_i^+ - x_i^-,\ x_i^+, x_i^- \ge 0$

LP / Simplex

(Basic Feasible Solution, BFS) $\boldsymbol{A}$ m $\boldsymbol{B}$ 0 $\boldsymbol{B}\boldsymbol{x}_B = \boldsymbol{b}$

1. BFS
2. (reduced cost) $\bar{\boldsymbol{c}}_N^\top = \boldsymbol{c}_N^\top - \boldsymbol{c}_B^\top \boldsymbol{B}^{-1} \boldsymbol{N}$
3. $\bar{\boldsymbol{c}}_N \geq \mathbf{0}$
4. Dantzig (ratio test)
5.

(Simplex Tableau)

LP

	$\boldsymbol{x}_B$	$\boldsymbol{x}_N$	RHS
$\boldsymbol{x}_B$	$\boldsymbol{I}$	$\boldsymbol{B}^{-1}\boldsymbol{N}$	$\boldsymbol{B}^{-1}\boldsymbol{b}$
z	$\mathbf{0}$	$\bar{\boldsymbol{c}}_N^\top$	$\boldsymbol{c}_B^\top \boldsymbol{B}^{-1}\boldsymbol{b}$

- $\bar{\boldsymbol{c}}_N \geq \mathbf{0}$
- $\bar{c}_j < 0$ $\leq \mathbf{0}$
- Min ratio test

(Two-Phase Method)

BFS

- **Phase I** $\boldsymbol{a}$

$$\min \sum_i a_i \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{x} + \boldsymbol{a} = \boldsymbol{b}, \;\; \boldsymbol{x}, \boldsymbol{a} \geq \mathbf{0}$$

> 0

- **Phase II** Phase I BFS
- M $M \sum a_i$

(Primal)

$$\min_{\boldsymbol{x}} \ \boldsymbol{c}^\top \boldsymbol{x} \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \ \boldsymbol{x} \geq \mathbf{0}$$

(Dual)

$$\max_{\boldsymbol{y}} \ \boldsymbol{b}^\top \boldsymbol{y} \quad \text{s.t.} \quad \boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}$$

Primal ( )	Dual ( )
$x_i \geq 0$	$\boldsymbol{a}_i^\top \boldsymbol{y} \leq c_i$
x_i	$\boldsymbol{a}_i^\top \boldsymbol{y} = c_i$
$\boldsymbol{a}_i^\top \boldsymbol{x} \geq b_i$	$y_i \geq 0$
$\boldsymbol{a}_i^\top \boldsymbol{x} = b_i$	y_i

(Weak Duality)

$\boldsymbol{x}$ Primal $\boldsymbol{y}$ Dual

$$\boldsymbol{c}^\top \boldsymbol{x} \geq \boldsymbol{b}^\top \boldsymbol{y}$$

(Strong Duality)

Primal Dual

$$\min \boldsymbol{c}^\top \boldsymbol{x} = \max \boldsymbol{b}^\top \boldsymbol{y}$$

(Complementary Slackness)

$x_i^* \cdot (\boldsymbol{a}_i^\top \boldsymbol{y}^* - c_i) = 0$ i

$y_i^* \cdot (\boldsymbol{a}_i^\top \boldsymbol{x}^* - b_i) = 0$ i

- $\boldsymbol{y}$ **(Shadow Price)**
- $y_i =$ i b_i 1
- 0
- > 0

(Sensitivity Analysis) b_i c_j

- $\boldsymbol{B}^{-1}\boldsymbol{b} \geq \mathbf{0}$
- $\bar{\boldsymbol{c}}_N \geq \mathbf{0}$

LP **(Interior Point Method)**

- KKT

$$
\begin{aligned}
\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, &\quad \boldsymbol{x} > \mathbf{0} \\
\boldsymbol{A}^\top \boldsymbol{y} + \boldsymbol{s} = \boldsymbol{c}, &\quad \boldsymbol{s} > \mathbf{0} \\
\boldsymbol{X}\boldsymbol{S}\boldsymbol{e} = \mu \boldsymbol{e} &
\end{aligned}
$$

$\boldsymbol{X} = \mathrm{diag}(\boldsymbol{x}),\ \boldsymbol{S} = \mathrm{diag}(\boldsymbol{s})$ $\mu \to 0$

Newton μ LP

LP

-
- m n
- Hungarian n n
- / $\rightarrow$ LP
-
- **(DEA)**

LP Gurobi, CPLEX, HiGHS, GLPK

NLP

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) \\ \text{s.t.} \quad & g_i(\boldsymbol{x}) \leq 0, \quad i = 1, \ldots, m \\ & h_j(\boldsymbol{x}) = 0, \quad j = 1, \ldots, p \end{aligned}$$

f, g_i, h_j

- **NLP** $\min f(\boldsymbol{x})$
- **NLP**
- $\rightarrow$

$\min f(\boldsymbol{x}), \ \boldsymbol{x} \in \mathbb{R}^n$

$\boldsymbol{x}^*$ f $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$

$\boldsymbol{x}^*$ f $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$ $\nabla^2 f(\boldsymbol{x}^*) \succeq \mathbf{0}$

$\nabla f(\boldsymbol{x}^*) = \mathbf{0}$ $\nabla^2 f(\boldsymbol{x}^*) \succ \mathbf{0}$ $\boldsymbol{x}^*$

(Gradient Descent)

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \alpha_k \nabla f(\boldsymbol{x}_k)$$

- α_k (step size / learning rate)
- $\mathcal{O}(1/k)$
- **Wolfe**
- **Backtracking line search**

ill-conditioned

(Newton's Method)

Taylor

$$f(\boldsymbol{x}_k + \boldsymbol{p}) \approx f(\boldsymbol{x}_k) + \nabla f(\boldsymbol{x}_k)^\top \boldsymbol{p} + \frac{1}{2}\boldsymbol{p}^\top \nabla^2 f(\boldsymbol{x}_k)\boldsymbol{p}$$

Newton $\boldsymbol{p}_k = -[\nabla^2 f(\boldsymbol{x}_k)]^{-1} \nabla f(\boldsymbol{x}_k)$

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k$$

-
-
- Hessian $\mathcal{O}(n^3)$ Hessian

(Quasi-Newton Methods)

Hessian Hessian

BFGS (Broyden–Fletcher–Goldfarb–Shanno)

$$
\begin{aligned}
s_k &= \boldsymbol{x}_{k+1} - \boldsymbol{x}_k \\
y_k &= \nabla f(\boldsymbol{x}_{k+1}) - \nabla f(\boldsymbol{x}_k) \\
\boldsymbol{H}_{k+1} &= \left(\boldsymbol{I} - \rho_k s_k y_k^\top\right) \boldsymbol{H}_k \left(\boldsymbol{I} - \rho_k y_k s_k^\top\right) + \rho_k s_k s_k^\top
\end{aligned}
$$

$\rho_k = 1/(y_k^\top s_k)$

-
- **L-BFGS** m (s_k, y_k) $\mathcal{O}(mn)$

Karush–Kuhn–Tucker (KKT)

NLP ——KKT

$$\nabla f(\boldsymbol{x}^*) + \sum_{i=1}^{m} \mu_i \nabla g_i(\boldsymbol{x}^*) + \sum_{j=1}^{p} \lambda_j \nabla h_j(\boldsymbol{x}^*) = \mathbf{0}$$

$$\begin{aligned} g_i(\boldsymbol{x}^*) &\leq 0, \quad i = 1, \ldots, m \\ h_j(\boldsymbol{x}^*) &= 0, \quad j = 1, \ldots, p \\ \mu_i &\geq 0, \quad i = 1, \ldots, m \\ \mu_i \cdot g_i(\boldsymbol{x}^*) &= 0, \quad i = 1, \ldots, m \quad (\quad) \end{aligned}$$

- μ_i Lagrange
- λ_j Lagrange
- KKT

- **(Penalty Method)**

$$\min_{\boldsymbol{x}} f(\boldsymbol{x}) + \rho \left[\sum_i \max(0, g_i(\boldsymbol{x}))^2 + \sum_j h_j(\boldsymbol{x})^2 \right]$$

$\rho \to \infty$ ill-conditioning

- **(Barrier Method)**

$$\min_{\boldsymbol{x}} f(\boldsymbol{x}) - \mu \sum_i \ln(-g_i(\boldsymbol{x}))$$

$\mu \to 0$

- **SQP (Sequential Quadratic Programming)** (QP)
- **Lagrange (Augmented Lagrangian)**

(Trust Region Methods)

Δ_k

$$\min_{\|\boldsymbol{p}\| \leq \Delta_k} m_k(\boldsymbol{p}) = f(\boldsymbol{x}_k) + \nabla f(\boldsymbol{x}_k)^\top \boldsymbol{p} + \frac{1}{2}\boldsymbol{p}^\top \boldsymbol{B}_k \boldsymbol{p}$$

ρ_k Δ_k

$$\rho_k = \frac{f(\boldsymbol{x}_k) - f(\boldsymbol{x}_k + \boldsymbol{p}_k)}{m_k(\mathbf{0}) - m_k(\boldsymbol{p}_k)}$$

-
- ill-conditioned
- Newton

NLP NP-hard

(Heuristic)

(Metaheuristic)

exploitation ( ) exploration ( )

- /
- (SA)
- (TS)
- (VNS)

- (GA)
- (PSO)
- (ACO)
- (DE)

(Greedy)

(Local Search) / (Hill Climbing)

- $N(\boldsymbol{x})$
- $f(\boldsymbol{x}') < f(\boldsymbol{x})$ $\boldsymbol{x}'$
- ______

" " " "

(SA)

$\rightarrow$ $\rightarrow$ $\rightarrow$

Metropolis

- $\Delta E = f(\boldsymbol{x}_{\text{new}}) - f(\boldsymbol{x}) < 0$
- $\Delta E \geq 0$ $p = \exp(-\Delta E / T)$

T cooling schedule T

- T_0
- α $T_{k+1} = \alpha \cdot T_k,\ \alpha \in (0.9, 0.99)$
- $T < T_{\min}$

1

(GA)

1. (population)
2. $f(\boldsymbol{x})$
3. **(Selection)** / ——
4. **(Crossover)**
5. **(Mutation)**
6. 2–5

TSP

- $p_c \approx 0.7$–0.9 $p_m \approx 0.001$–0.1
- (Elitism)

(PSO)

i

- $\boldsymbol{x}_i$
- $\boldsymbol{v}_i$
- $\boldsymbol{p}_{\text{best},i}$
- $\boldsymbol{g}_{\text{best}}$

$$\boldsymbol{v}_i \leftarrow \omega \boldsymbol{v}_i + c_1 r_1 (\boldsymbol{p}_{\text{best},i} - \boldsymbol{x}_i) + c_2 r_2 (\boldsymbol{g}_{\text{best}} - \boldsymbol{x}_i)$$
$$\boldsymbol{x}_i \leftarrow \boldsymbol{x}_i + \boldsymbol{v}_i$$

- ω vs
- c_1, c_2 / ≈ 2
-

(Tabu Search)

(Tabu List)

- “ ”
- (Tabu Tenure) ℓ
- (Aspiration Criterion)
- (Long-term Memory)

TSP

(Ant Colony Optimization, ACO)

(pheromone)

TSP ACO

- τ_{ij} (i, j)
- k i j

$$p_{ij}^k = \frac{\tau_{ij}^\alpha \cdot \eta_{ij}^\beta}{\sum_{\ell \in N_i^k} \tau_{i\ell}^\alpha \cdot \eta_{i\ell}^\beta}$$

  $\eta_{ij} = 1/d_{ij}$ /

- (Evaporation) $\tau_{ij} \leftarrow (1-\rho)\tau_{ij}$
-

SA	$+$		$/$
GA	$+$		$/$
PSO	$+$	$\boldsymbol{p}_{\text{best}}, \boldsymbol{g}_{\text{best}}$	
TS	$+$		
ACO	$+$		$/$

No Free Lunch Theorem

(Convex Sets)

$C \subseteq \mathbb{R}^n$ $\quad$ $\boldsymbol{x}, \boldsymbol{y} \in C \quad \theta \in [0, 1]$

$$\theta \boldsymbol{x} + (1 - \theta) \boldsymbol{y} \in C$$

- $\{\boldsymbol{x} \mid \boldsymbol{a}^\top \boldsymbol{x} = b\}$
- $\{\boldsymbol{x} \mid \boldsymbol{a}^\top \boldsymbol{x} \leq b\}$
- $\{\boldsymbol{x} \mid \boldsymbol{A}\boldsymbol{x} \preceq \boldsymbol{b}\}$
-
- $\{(\boldsymbol{x}, t) \mid \|\boldsymbol{x}\| \leq t\}$

- $\bigcap_\alpha C_\alpha$
- $f(C)$
- /

(Convex Functions)

$f : \mathbb{R}^n \to \mathbb{R}$ $\boldsymbol{x}, \boldsymbol{y}$ $\theta \in [0, 1]$

$$f(\theta \boldsymbol{x} + (1-\theta)\boldsymbol{y}) \leq \theta f(\boldsymbol{x}) + (1-\theta) f(\boldsymbol{y})$$

- $f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\top (\boldsymbol{y} - \boldsymbol{x})$
- $\nabla^2 f(\boldsymbol{x}) \succeq \mathbf{0}$ Hessian
- (Epigraph) $\mathrm{epi}(f) = \{(\boldsymbol{x}, t) \mid f(\boldsymbol{x}) \leq t\}$ $\iff$ f

$f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\top (\boldsymbol{y} - \boldsymbol{x}) + \frac{\mu}{2}\|\boldsymbol{y} - \boldsymbol{x}\|^2$ $\mu > 0$

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & f_0(\boldsymbol{x}) \\ \text{s.t.} \quad & f_i(\boldsymbol{x}) \leq 0, \quad i = 1, \ldots, m \\ & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \end{aligned}$$

$f_0, f_1, \ldots, f_m$

= KKT

- LP, QP ( ), QCQP ( )
- SOCP ( ), SDP ( )
- (Geometric Programming)

Lagrange

$$g(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{\boldsymbol{x}} \mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$$

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f_0(\boldsymbol{x}) + \sum_i \lambda_i f_i(\boldsymbol{x}) + \sum_j \nu_j(\boldsymbol{a}_j^\top \boldsymbol{x} - b_j)$$

Slater (Slater's Condition)

$$\text{Primal} = \text{Dual}$$

(Subgradient Method)

f

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \alpha_k \boldsymbol{g}_k, \quad \boldsymbol{g}_k \in \partial f(\boldsymbol{x}_k)$$

$\partial f(\boldsymbol{x}) =$

- $\alpha_k \to 0, \ \sum \alpha_k = \infty$
- $\mathcal{O}(1/\sqrt{k})$
- ℓ_1 SVM

(Proximal Gradient Method)

$\min f(\boldsymbol{x}) + g(\boldsymbol{x})$ f g prox

(Proximal Operator)

$$\operatorname{prox}_{\eta g}(\boldsymbol{v}) = \arg\min_{\boldsymbol{x}} \left\{ g(\boldsymbol{x}) + \frac{1}{2\eta} \|\boldsymbol{x} - \boldsymbol{v}\|^2 \right\}$$

ISTA / FISTA /

$$\begin{aligned} \boldsymbol{z}_k &= \boldsymbol{x}_k - \eta \nabla f(\boldsymbol{x}_k) \\ \boldsymbol{x}_{k+1} &= \operatorname{prox}_{\eta g}(\boldsymbol{z}_k) \end{aligned}$$

- ℓ_1 (LASSO)
- FISTA $\mathcal{O}(1/k^2)$

Nesterov (Accelerated Gradient)

$\mathcal{O}(1/k)$ Nesterov $\mathcal{O}(1/k^2)$

Nesterov (NAG)

$$\boldsymbol{y}_k = \boldsymbol{x}_k + \beta_k(\boldsymbol{x}_k - \boldsymbol{x}_{k-1})$$
$$\boldsymbol{x}_{k+1} = \boldsymbol{y}_k - \alpha_k \nabla f(\boldsymbol{y}_k)$$

$\beta_k = \frac{k-1}{k+2}$

- "Look ahead"
- $\mathcal{O}\left((1 - \sqrt{\mu/L})^k\right)$
- Nesterov SGD with Momentum Adam

ADMM (Alternating Direction Method of Multipliers)

$$\begin{aligned} \min_{\boldsymbol{x},\boldsymbol{z}} \quad & f(\boldsymbol{x}) + g(\boldsymbol{z}) \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} = \boldsymbol{c} \end{aligned}$$

Lagrange $\mathcal{L}_\rho(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{y}) = f(\boldsymbol{x}) + g(\boldsymbol{z}) + \boldsymbol{y}^\top(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c}) + \frac{\rho}{2}\|\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c}\|^2$

ADMM

$$\begin{aligned} \boldsymbol{x}_{k+1} &= \arg\min_{\boldsymbol{x}} \mathcal{L}_\rho(\boldsymbol{x}, \boldsymbol{z}_k, \boldsymbol{y}_k) \\ \boldsymbol{z}_{k+1} &= \arg\min_{\boldsymbol{z}} \mathcal{L}_\rho(\boldsymbol{x}_{k+1}, \boldsymbol{z}, \boldsymbol{y}_k) \\ \boldsymbol{y}_{k+1} &= \boldsymbol{y}_k + \rho(\boldsymbol{A}\boldsymbol{x}_{k+1} + \boldsymbol{B}\boldsymbol{z}_{k+1} - \boldsymbol{c}) \end{aligned}$$

(SDP)

SDP LP

$$\begin{aligned} \min_{\boldsymbol{X}} \quad & \langle \boldsymbol{C}, \boldsymbol{X} \rangle = \operatorname{tr}(\boldsymbol{C}^\top \boldsymbol{X}) \\ \text{s.t.} \quad & \langle \boldsymbol{A}_i, \boldsymbol{X} \rangle = b_i, \quad i = 1, \ldots, m \\ & \boldsymbol{X} \succeq \mathbf{0} \quad (\quad) \end{aligned}$$

- (Primal-Dual IPM)
- PCA SDP
- Max-Cut Goemans–Williamson 0.878 SDP

LP $\subset$ SOCP $\subset$ SDP

- CVXPY (Python)
- CVX (MATLAB)
- Convex.jl (Julia)

- ECOS, SCS —SOCP, SDP
- OSQP —QP
- MOSEK — LP/QP/SOCP/SDP
- Gurobi / CPLEX —LP, MILP, QP

- LP → LP
- QP → QP
- SOCP/SDP →
- NLP (IPOPT, SNOPT)

=

$$\min_{\boldsymbol{\theta}} \frac{1}{N} \sum_{i=1}^{N} \ell(\boldsymbol{x}_i, y_i;\ \boldsymbol{\theta}) + \lambda R(\boldsymbol{\theta})$$

- $\ell(\cdot)$ MSE Hinge
- $R(\cdot)$ ℓ_2, ℓ_1, dropout
- N ML

ML

- N $\rightarrow$
- n $\rightarrow$
-
- $\gg$

(SGD)

(mini-batch) $\mathcal{B}$

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta_t \cdot \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \nabla_\theta \ell(\boldsymbol{x}_i, y_i;\ \boldsymbol{\theta}_t)$$

- $\mathbb{E}[\nabla f_{\mathcal{B}}(\boldsymbol{\theta})] = \nabla f(\boldsymbol{\theta})$
- $\mathcal{O}(1/\sqrt{T})$ $\mathcal{O}(1/T)$
- batch size

SGD ill-conditioned

(Momentum) Nesterov

SGD with Momentum (Polyak)

$$\begin{aligned} \boldsymbol{v}_t &= \beta \boldsymbol{v}_{t-1} + \eta \nabla f(\boldsymbol{\theta}_t) \\ \boldsymbol{\theta}_{t+1} &= \boldsymbol{\theta}_t - \boldsymbol{v}_t \end{aligned}$$

Nesterov Accelerated Gradient (NAG)

$$\begin{aligned} \boldsymbol{v}_t &= \beta \boldsymbol{v}_{t-1} + \eta \nabla f(\boldsymbol{\theta}_t - \beta \boldsymbol{v}_{t-1}) \\ \boldsymbol{\theta}_{t+1} &= \boldsymbol{\theta}_t - \boldsymbol{v}_t \end{aligned}$$

-
- NAG "Look-ahead"
- $\beta = 0.9$

Adam (Adaptive Moment Estimation)

$$
\begin{aligned}
\boldsymbol{m}_t &= \beta_1 \boldsymbol{m}_{t-1} + (1-\beta_1)\boldsymbol{g}_t \\
\boldsymbol{v}_t &= \beta_2 \boldsymbol{v}_{t-1} + (1-\beta_2)\boldsymbol{g}_t^2 \\
\hat{\boldsymbol{m}}_t &= \boldsymbol{m}_t/(1-\beta_1^t), \quad \hat{\boldsymbol{v}}_t = \boldsymbol{v}_t/(1-\beta_2^t) \\
\boldsymbol{\theta}_{t+1} &= \boldsymbol{\theta}_t - \eta \cdot \frac{\hat{\boldsymbol{m}}_t}{\sqrt{\hat{\boldsymbol{v}}_t} + \epsilon}
\end{aligned}
$$

- $\hat{\boldsymbol{m}}_t$
- $\hat{\boldsymbol{v}}_t$ RMS
- $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$
- AdamW weight decay AdaBelief LAMB

(Learning Rate Schedule)

η

- **Step decay** epoch
- **Cosine annealing**

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min})\left(1 + \cos\frac{t\pi}{T}\right)$$

- **Warmup** epoch η Transformer
- **Cyclic / OneCycle** LR

Warmup

Transformer Warmup warmup Adam

- **Natural Gradient / K-FAC** Fisher Hessian
- **Shampoo** Google
- **AdaHessian** Hutchinson Hessian
- **Newton Sketch** Newton

Hessian $n \times n$ $\rightarrow$ Hessian

> 100B GPU

(Data Parallel) $+$ **All-Reduce**

- GPU mini-batch
- All-Reduce

(Model Parallel)

- Tensor Parallelism GPU
- Pipeline Parallelism GPU

- **Gradient Accumulation** batch
- **(FP16/BF16)**
- **ZeRO** (Zero Redundancy Optimizer)

Microsoft DeepSpeed ZeRO GPU

ZeRO-1	(Adam m, v)	4$\times$
ZeRO-2	$+$	8$\times$
ZeRO-3	$+$ $+$	GPU

175B GPT-3 GPU

LLM

(Full Fine-tuning)

-
- 16-32 $+$ $+$

(PEFT)

- **LoRA** $\Delta W = BA$
- **Adapter**
- **Prefix / Prompt Tuning** token

RLHF

RLHF (Reinforcement Learning from Human Feedback)

1. SFT (Supervised Fine-Tuning)
2. Reward Model Bradley-Terry
3. PPO (Proximal Policy Optimization) Reward Model

$$\max_{\theta} \mathbb{E}\left[r(x, y) - \beta \cdot \mathsf{KL}(\pi_\theta \parallel \pi_{\text{ref}})\right]$$

DPO (Direct Preference Optimization)—— RL reward PPO

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}\left[\log \sigma\left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\right)\right]$$

- **(GPTQ, AWQ)** FP16 4-bit 8-bit
  - (OBS)
  - $\min_{\hat{W}} \| WX - \hat{W}X \|_F^2$
- **KV Cache** Key-Value
- **(Speculative Decoding)** token
- **FlashAttention** IO-aware

Pareto

LLM

- **(Loss Spike)**
- FP16/BF16 underflow / overflow
- **(Gradient Clipping)** $\|\boldsymbol{g}\| > \tau$ $\boldsymbol{g} \leftarrow \tau \cdot \boldsymbol{g}/\|\boldsymbol{g}\|$
- **Z-loss / Logit Regularization** logit
- output embedding LayerNorm

Chinchilla Scaling Law C N token D

$$N_{\text{opt}} \propto C^{0.5}, \quad D_{\text{opt}} \propto C^{0.5}$$

ML

- **Decision-Focused Learning** ML
  - $\rightarrow$ LP/IP
- **Learning to Optimize** RL
- Pointer Network / Transformer TSP, VRP
- **+ ML** GNN MILP
- **Convex ML** $+$ Elastic Net
- **Bilevel Optimization**

GPU

ML —— GPU SIMT

CPU ( )

- 8–64
-
-

GPU ( )

-
- (SIMT)
- (~2 TB/s)

NVIDIA CUDA

- Thread → Warp (32 threads) → Block → Grid
- Global → L2 → Shared Memory → Registers
- Tensor Cores (Volta+) 4×4

A100 80GB: 6912 CUDA cores + 432 Tensor Cores, 312 TFLOPS (FP16)

GPU

(SGD, Adam, AdamW)

1. Forward pass $\boldsymbol{y} = \boldsymbol{W}\boldsymbol{x}$
2. Loss
3. Backward pass $\nabla_W \mathcal{L} = (\nabla_y \mathcal{L})\boldsymbol{x}^\top$
4.

1 3 GPU $50\times$–$200\times$

- cuBLAS GEMM ( )
- cuDNN GPU kernel
- (Newton, K-FAC) Hessian GPU

GPU

175B ∼350 GB (FP16) TB

(Mixed Precision)

FP32 FP16

(NVIDIA APEX / PyTorch AMP)

1. FP32
2. Forward/Backward FP16 $2\times$
3. **Loss Scaling** S underflow
4. FP32

BF16 (Brain Float 16)

- FP32 (8 bit) (7 bit)
- $=$ FP32 loss scaling
- NVIDIA A100 / H100

Tensor Cores FP16/BF16 FP32 8–16$\times$

&

GPU batch size

(Gradient Accumulation)

- k micro-batch $+$
- k batch size $\times k$
- $=$ micro-batch size $\approx$

- (Linear Scaling Rule, Goyal et al. 2017)
  $\eta_{\text{new}} = \eta_{\text{base}} \times (B_{\text{new}}/B_{\text{base}})$ warmup
- LARS (Layer-wise Adaptive Rate Scaling)
- LAMB Adam BERT

GPU

(Data Parallel)

- GPU
- GPU mini-batch
- All-Reduce
- PyTorch DDP, Horovod

- Tensor Parallel (Megatron-LM)
- Pipeline Parallel (GPipe)
- 3D Parallel = DP + TP + PP

NCCL

- All-Reduce
- All-Gather
- Reduce-Scatter
- ∼600 GB/s (NVLink)

FSDP (Fully Sharded DP)

\+ + ZeRO-3 PyTorch

1B + Adam + FP16

(FP16)	2 GB	∼20%
Adam m (FP32)	4 GB	∼40%
Adam v (FP32)	4 GB	∼40%
(FP16)	2 GB	∼20%
(Activations)		

- **Gradient Checkpointing** $\searrow$ $\nearrow$
- **Activation Offloading** CPU
- **FlashAttention** IO-aware + softmax $\mathcal{O}(N^2)$ $\mathcal{O}(N)$
- **Kernel Fusion** CUDA kernel Bias + GeLU + Dropout

GPU Checklist

1. AMP (FP16) BF16 Tensor Core
2. DataLoader prefetch pin_memory
3. batch + batch
4. DDP FSDP / DeepSpeed
5. torch.compile / TensorRT XLA kernel fusion
6. gradient checkpointing FlashAttention fused optimizer
7. nvidia-smi, Nsight Systems (timeline), Nsight Compute (kernel)
8. **batch size** GPU ($> 80\%$)

GPU ——

GPU

- **LP** GPU cuSOLVER
- MIP B&B GPU
- Transformer/GNN GPU TSP/VRP
- GPU
- **RL for OR** GPU PPO /
- **SDP** GPU SDP Cholesky

NVIDIA cuOpt GPU VRP $100\times$

1.
2. KKT /
3. ADMM
4. **ML** SGD $\rightarrow$ Momentum $\rightarrow$ Adam
5. **LLM** ZeRO PEFT LoRA RLHF
6. ML OR
7. **GPU** CUDA FlashAttention cuOpt